Project Design Phase-I

Solution Architecture

Date	08 March 2026
Team ID	—
Project Name	Predictive Pulse – Hypertension Risk Detection using Machine Learning
Maximum Marks	4 Marks

Architecture Layers

#	Layer	Component	Technology & Role
1	Frontend (UI)	Web Interface	**HTML5 + CSS3** — Provides a medical dashboard where users enter health details such as age group, symptoms, blood pressure levels, and medical history. Displays prediction results and recommendations.
2	Web Server	Flask Application (app.py)	**Python Flask Framework** — Handles HTTP requests, processes form data from the frontend, and connects the user interface with the machine learning model.
3	Processing Layer	Data Preprocessing	Converts user input into numerical format, encodes categorical variables, and scales input features before sending them to the prediction model.
4	Machine Learning Layer	Hypertension Prediction Model	**Logistic Regression Model (Scikit-learn)** — Predicts the hypertension stage based on user health parameters and symptoms.
5	Model Storage	Trained Model File	**logreg_model.pkl** — Saved machine learning model loaded by Flask using Joblib to perform predictions in real time.
6	Recommendation Layer	Health Advice System	Generates **medical recommendations and lifestyle advice** based on predicted hypertension stage and risk level.
7	Data Layer	Input Features	Patient health parameters such as **age group, gender, symptoms, blood pressure readings, medication history, and diet control** used by the ML model for prediction.
8	Output Layer	Result Display	Shows **predicted hypertension stage, risk level color, confidence score, and preventive health recommendations** on the web interface.

Data / Request Flow

Step	Action	Component
1	User enters health details through the web interface	Browser / HTML Form (templates/index.html)
2	Flask server receives the form submission	Flask Web Server (app.py)
3	Input data is validated and preprocessed	Data Processing Layer
4	Processed input features are passed to the ML model	Logistic Regression Model
5	The model predicts the hypertension stage	Machine Learning Layer
6	Confidence score and health recommendations are generated	Recommendation System
7	Results are sent back to the frontend	Flask → Browser
8	User sees prediction results and medical advice	Web Interface